Series Solutions of ODEs. Special Functions

5.1 Power Series Method

The **power series method** is the standard method for solving linear ODEs with ***variable*** coefficients. It gives solutions in the form of power series. These series can be used for computing values, graphing curves, proving formulas, and exploring properties of solutions, as we shall see. In this section we begin by explaining the idea of the power series method.

From calculus we remember that a **power series** (in powers of $x - x_0$) is an infinite series of the form

$$\sum_{m=0}^{\infty} a_m(x - x_0)^m = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \cdots. \tag{1}$$

Here, x is a variable. $a_0, a_1, a_2, \cdots$ are constants, called the **coefficients** of the series. x_0 is a constant, called the **center** of the series. In particular, if $x_0 = 0$, we obtain a **power series in powers of x**

$$\sum_{m=0}^{\infty} a_m x^m = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots. \tag{2}$$

Idea and Technique of the Power Series Method

EXAMPLE 2 **Power Series Solution.** Solve $y' - y = 0$.

Solution. In the first step we insert

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots = \sum_{m=0}^{\infty} a_m x^m \tag{2}$$

$y' =$

and the series obtained by termwise differentiation

(3)
$$y' = a_1 + 2a_2x + 3a_3x^2 + \cdots = \sum_{m=1}^{\infty} ma_m x^{m-1}$$

into the ODE:

$$(a_1 + 2a_2x + 3a_3x^2 + \cdots) - (a_0 + a_1x + a_2x^2 + \cdots) = 0.$$

Then we collect like powers of x, finding

$$(a_1 - a_0) + (2a_2 - a_1)x + (3a_3 - a_2)x^2 + \cdots = 0.$$

Equating the coefficient of each power of x to zero, we have

$$a_1 - a_0 = 0, \qquad 2a_2 - a_1 = 0, \qquad 3a_3 - a_2 = 0, \cdots.$$

Solving these equations, we may express $a_1, a_2, \cdots$ in terms of a_0, which remains arbitrary:

$$a_1 = a_0, \qquad a_2 = \frac{a_1}{2} = \frac{a_0}{2!}, \qquad a_3 = \frac{a_2}{3} = \frac{a_0}{3!}, \cdots.$$

With these values of the coefficients, the series solution becomes the familiar general solution

$$y = a_0 + a_0x + \frac{a_0}{2!}x^2 + \frac{a_0}{3!}x^3 + \cdots = a_0\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}\right) = a_0e^x.$$

Thus, the solution is given by

$$y = a_0e^x$$

The above problem can do in the following way:

Q.) Solve $y' - y = 0$ $\longrightarrow$ (1)

Let $y = \sum_{m=0}^{\infty} a_m x^m$ $\quad y' = \sum_{m=1}^{\infty} m a_m x^{m-1}$

we can write y' as,

$$y' = \sum_{m=0}^{\infty} (m+1)\, a_{m+1}\, x^m \quad (m \to m+1)$$

Then (1) becomes

$$\sum_{m=0}^{\infty} (m+1)\, a_{m+1}\, x^m - \sum_{m=0}^{\infty} a_m x^m = 0$$

Combining,

$$\sum_{m=0}^{\infty} \left[(m+1)\, a_{m+1} - a_m\right] x^m = 0$$

$\Rightarrow (m+1)\, a_{m+1} - a_m = 0$

$\Rightarrow$ $\boxed{a_{m+1} = \frac{a_m}{m+1}}$ recurrence

Put $m=0$, $\quad a_1 = a_0$

$m=1$ $\quad a_2 = \frac{a_1}{2} = \frac{a_1}{2!} = \frac{a_0}{2!}$

$m=2$ $\quad a_3 = \frac{a_2}{3} = \frac{a_0}{2! \times 3} = \frac{a_0}{3!}$

$m=3$ $\quad a_4 = \frac{a_3}{4} = \frac{a_0}{3! \times 4} = \frac{a_0}{4!}$

$\therefore y = \sum_{m=0}^{\infty} a_m x^m$

$= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$

$= a_0 + a_0 x + \frac{a_0}{2!} x^2 + \frac{a_0}{3!} x^3 + \cdots$

$= a_0 \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right) = a_0 e^x$

NOTE:

$$y'' = 2a_2 + 3 \cdot 2a_3 x + 4 \cdot 3a_4 x^2 + \cdots = \sum_{m=2}^{\infty} m(m-1) a_m x^{m-2}$$

$$= \sum_{m=0}^{\infty} (m+2)(m+1) a_{m+2} x^m$$

2) Solve $y'' + y = 0 \longrightarrow (1)$

Soln let $y = \sum_{m=0}^{\infty} a_m x^m$

$$y' = \sum_{m=1}^{\infty} m\, a_m x^{m-1} = \sum_{m=0}^{\infty} (m+1)\, a_{m+1} x^m$$

$$y'' = \sum_{m=2}^{\infty} m(m-1)\, a_m x^{m-2}$$

$$= \sum_{m=0}^{\infty} (m+2)(m+1)\, a_{m+2} x^m$$

Substituting in eqn (1) $y'' + y = 0$

$$\sum_{m=0}^{\infty} (m+2)(m+1)\, a_{m+2} x^m + \sum_{m=0}^{\infty} a_m x^m = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} \left[(m+2)(m+1)\, a_{m+2} + a_m\right] x^m = 0$$

$$\Rightarrow (m+2)(m+1)\, a_{m+2} + a_m = 0$$

The recurrence relation

$$a_{m+2} = \frac{-a_m}{(m+1)(m+2)}$$

✓

Let m € even) including $m=0$

when $m=0$

$$a_2 = \frac{-a_0}{2} = \frac{-a_0}{2!}$$

when $m=2$

$$a_4 = \frac{-a_2}{3 \cdot 4} = \frac{a_0}{2! \times 3 \times 4} = \frac{a_0}{4!}$$

when $m=4$

$$a_6 = \frac{-a_4}{5\times 6} = \frac{-a_0}{4!\times 5\times 6} = \frac{-a_0}{6!}$$

$$a_{2m} = \frac{(-1)^m a_0}{(2m)!}$$

Let m (odd)

when $m=1$

$$a_3 = \frac{-a_1}{2\times 3} = \frac{-a_1}{3!}$$

when $m=3$

$$a_5 = \frac{-a_3}{4\times 5} = \frac{a_1}{3!\times 4\times 5} = \frac{a_1}{5!}$$

when $m=5$

$$a_7 = \frac{-a_5}{6\times 7} = \frac{-a_1}{5!\times 6\times 7} = \frac{-a_1}{7!}$$

$$a_{2m+1} = \frac{(-1)^m a_1}{(2m+1)!}$$

Final solution:

$$y = \sum_{m=0}^{\infty} a_m x^m$$

$$= a_0 \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} x^{2m} + a_1 \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} x^{2m+1}$$

2m+1

we know that

$$\sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} x^{2m} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots = \cos x$$

$$\sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} x^{2m+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = \sin x$$

∴ the solution is, $y = a_0 \cos x + a_1 \sin x$

Q:) Find a power series solution in powers of x.

$$y'' - y' + x^2 y = 0$$

Solⁿ

$$\sum_{m=0}^{\infty} (m+2)(m+1)a_{m+2}x^m - \sum_{m=0}^{\infty}(m+1)a_{m+1}x^m + x^2\sum_{m=0}^{\infty} a_m x^m = 0 \quad \longrightarrow (1)$$

Third term

$$x^2 \sum_{m=0}^{\infty} a_m x^m = a_0 x^2 + a_1 x^3 + a_2 x^4 + \cdots$$

$$= \sum_{m=2}^{\infty} a_{m-2} x^m \quad \left(\begin{array}{l} m=0 \\ m=1 \\ \text{take value is} \\ \text{zero} \end{array}\right)$$

$$= \sum_{m=0}^{\infty} a_{m-2} x^m$$

(1) becomes

$$\sum_{m=0}^{\infty} \left[(m+2)(m+1)a_{m+2} - (m+1)a_{m+1} + a_{m-2}\right] x^m = 0$$

$$\Rightarrow (m+2)(m+1)a_{m+2} - (m+1)a_{m+1} + a_{m-2} = 0$$

$$\therefore \boxed{a_{m+2} = \frac{(m+1)a_{m+1} - a_{m-2}}{(m+1)(m+2)}}$$

$m=0$ $\quad a_2 = \frac{a_1}{2}$

$m=1$ $\quad a_3 = \frac{2a_2}{6} = \frac{2 \times a_1/2}{6} = \frac{a_1}{6}$

$m=2$ $\quad a_4 = \frac{3a_3 - a_0}{12} = \frac{\frac{3a_1}{6} - a_0}{12} = \frac{a_1 - 2a_0}{24}$

$m = 3$

$$a_5 = \frac{4a_4 - a_1}{20}$$

$$= \frac{4\left(\frac{a_1 - 2a_0}{24}\right) - a_1}{20}$$

$$= \frac{a_1 - 2a_0 - 6a_1}{120}$$

$$= \frac{-2a_0 - 5a_1}{120}$$ ✓

∴ The solution is,

$$y = \sum_{m=0}^{\infty} a_m x^m$$

$$= a_0 + a_1 x + \frac{a_1}{2} x^2 + \frac{a_1}{6} x^3 + \frac{(a_1 - 2a_0)}{24} x^4 + \frac{(-2a_0 - 5a_1)}{120} x^5 + \cdots$$ ✓

$$= \left(a_0 - \frac{a_0}{12} x^4 - \frac{a_0}{60} x^5 + \cdots\right) + \left(a_1 x + \frac{a_1}{2} x^2 + \frac{a_1}{6} x^3 + \frac{a_1}{24} x^4 - \frac{a_1}{24} x^5 - \cdots\right)$$ ✓

$$y = a_0\left(1 - \frac{x^4}{12} - \frac{x^5}{60} - \cdots\right) + a_1\left(x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} - \frac{x^5}{24} + \cdots\right)$$

Q:) Solve the initial value problem by a power series. Find the value of the partial sum S at x upto and including x^5

$$y'' + 3xy' + 2y = 0, \quad y(0) = 1, \quad y'(0) = 1 \qquad x = 0.5$$

$\longrightarrow$ (1)

Soln: Let $y = \sum_{m=0}^{\infty} a_m x^m$ ✓

(1) becomes

$$\sum_{m=0}^{\infty} (m+2)(m+1)\, a_{m+2} x^m + 3x \sum_{m=0}^{\infty} (m+1)\, a_{m+1} x^m + 2 \sum_{m=0}^{\infty} a_m x^m = 0 \longrightarrow (2)$$

2nd term

$$x \sum_{m=0}^{\infty} (m+1)\, a_{m+1} x^m = a_1 x + 2a_2 x^2 + 3a_3 x^3 + \cdots = \sum_{m=0}^{\infty} m\, a_m x^m$$ ✓

(2) becomes

$$\sum_{m=0}^{\infty} (m+2)(m+1)\, a_{m+2} x^m + 3 \sum_{m=0}^{\infty} m\, a_m x^m + 2 \sum_{m=0}^{\infty} a_m x^m = 0$$

Combining

$$\sum_{m=0}^{\infty} \left[(m+2)(m+1)\, a_{m+2} + 3m a_m + 2a_m \right] x^m = 0$$

$$\Rightarrow (m+2)(m+1)\, a_{m+2} + (3m+2)\, a_m = 0$$

$$\Rightarrow \boxed{a_{m+2} = \frac{-(3m+2)\, a_m}{(m+1)(m+2)}}$$ ✓

Even terms

$m=0$ $\quad a_2 = -\frac{2}{2} a_0 = -a_0$

$m=2$ $\quad a_4 = \frac{-8}{12} a_2 = \frac{2}{3} a_0$

$m=4$ $\quad a_6 = \frac{-14}{30} a_4 = \frac{-14}{45} a_0$

Odd terms

$m=1$ $\quad a_3 = -\frac{5}{6} a_1$

$m=3$ $\quad a_5 = \frac{-11}{20} a_3 = \frac{11}{24} a_1$

$m=5$ $\quad a_7 = \frac{-17}{42} a_5 = \frac{-187}{1008} a_1$

∴ Solution is given by

$$y = a_0 + a_1 x - a_0 x^2 - \frac{5}{6} a_1 x^3 + \frac{2}{3} a_0 x^4 + \frac{11}{24} a_1 x^5 + \dots$$

$$y = a_0 \left(1 - x^2 + \frac{2}{3} x^4 \dots\right) + a_1 \left(x - \frac{5}{6} x^3 + \frac{11}{24} x^5 - \dots\right)$$ ✓

now, upto and including x^5, we have

$$y = a_0 \left(1 - x^2 + \frac{2}{3} x^4\right) + a_1 \left(x - \frac{5}{6} x^3 + \frac{11}{24} x^5\right)$$ ✓

Apply the IVP

$y(0) = 1 \Rightarrow$

$$\boxed{1 = a_0}$$

Now $y' = a_0\left(-2x + \frac{8}{3}x^3\right) + a_1\left(1 - \frac{5}{2}x^2 + \frac{55}{24}x^4\right)$

$y'(0) = 1 \Rightarrow$ $\boxed{a_1 = 1}$

$1 = 0 + a_1 \Rightarrow$

$\therefore y = 1 - x^2 + \frac{2}{3}x^4 + x - \frac{5}{6}x^3 + \frac{11}{24}x^5$

$y = 1 + x - x^2 - \frac{5}{6}x^3 + \frac{2}{3}x^4 + \frac{11}{24}x^5$

At the point $x = 0.5$

$y(0.5) = 1.2018$

EXAMPLE 3 **A Special Legendre Equation.** The ODE

$$(1 - x^2)y'' - 2xy' + 2y = 0$$

occurs in models exhibiting spherical symmetry. Solve it.

$y'' - x^2y'' - 2xy' + 2y = 0$

Solution. Substitute (2), (3), and (5) into the ODE. $(1 - x^2)y''$ gives two series, one for y'' and one for $-x^2y''$. In the term $-2xy'$ use (3) and in $2y$ use (2). Write like powers of x vertically aligned. This gives

$$\begin{aligned}
y'' &= 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + 30a_6x^4 + \cdots \\
-x^2y'' &= \qquad\quad - 2a_2x^2 - 6a_3x^3 - 12a_4x^4 - \cdots \\
-2xy' &= \quad - 2a_1x - 4a_2x^2 - 6a_3x^3 - 8a_4x^4 - \cdots \\
2y &= 2a_0 + 2a_1x + 2a_2x^2 + 2a_3x^3 + 2a_4x^4 + \cdots.
\end{aligned}$$

Add terms of like powers of x. For each power $x^0, x, x^2, \cdots$ equate the sum obtained to zero. Denote these sums by [0] (constant terms), [1] (first power of x), and so on:

Sum	Power	Equations	
[0]	$[x^0]$	$a_2 = -a_0$	
[1]	$[x]$	$a_3 = 0$	
[2]	$[x^2]$	$12a_4 = 4a_2,$	$a_4 = \frac{4}{12}a_2 = -\frac{1}{3}a_0$
[3]	$[x^3]$	$a_5 = 0$	since $a_3 = 0$
[4]	$[x^4]$	$30a_6 = 18a_4,$	$a_6 = \frac{18}{30}a_4 = \frac{18}{30}(-\frac{1}{3})a_0 = -\frac{1}{5}a_0.$

This gives the solution

$$y = a_1 x + a_0(1 - x^2 - \tfrac{1}{3}x^4 - \tfrac{1}{5}x^6 - \cdots).$$

a_0 and a_1 remain arbitrary.

Theory of the Power Series Method

The ***n*th partial sum** of (1) is

$$(6) \qquad s_n(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \cdots + a_n(x - x_0)^n$$

where $n = 0, 1, \cdots$. If we omit the terms of s_n from (1), the remaining expression is

$$(7) \qquad R_n(x) = a_{n+1}(x - x_0)^{n+1} + a_{n+2}(x - x_0)^{n+2} + \cdots.$$

This expression is called the **remainder** *of* (1) *after the term* $a_n(x - x_0)^n$.

For example, in the case of the geometric series

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \cdots + x^n + \cdots$$

we have

$$s_0 = 1, \qquad R_0 = x + x^2 + x^3 + \cdots,$$

$$s_1 = 1 + x, \qquad R_1 = x^2 + x^3 + x^4 + \cdots,$$

$$s_2 = 1 + x + x^2, \qquad R_2 = x^3 + x^4 + x^5 + \cdots, \qquad \text{etc.}$$

In this way we have now associated with (1) the sequence of the partial sums $s_0(x), s_1(x), s_2(x), \cdots$. If for some $x = x_1$ this sequence converges, say,

$$\lim_{n\to\infty} s_n(x_1) = s(x_1),$$

then the series (1) is called **convergent** *at* $x = x_1$, the number $s(x_1)$ is called the **value** or *sum* of (1) at x_1, and we write

$$s(x_1) = \sum_{m=0}^{\infty} a_m(x_1 - x_0)^m.$$

Then we have for every n,

$$s(x_1) = s_n(x_1) + R_n(x_1). \tag{8}$$

If that sequence diverges at $x = x_1$, the series (1) is called **divergent** at $x = x_1$.

In the case of convergence, for any positive ϵ there is an N (depending on ϵ) such that, by (8)

$$|R_n(x_1)| = |s(x_1) - s_n(x_1)| < \epsilon \qquad \text{for all } n > N. \tag{9}$$

Geometrically, this means that all $s_n(x_1)$ with $n > N$ lie between $s(x_1) - \epsilon$ and $s(x_1) + \epsilon$ (Fig. 104). Practically, this means that in the case of convergence we can approximate the sum $s(x_1)$ of (1) at x_1 by $s_n(x_1)$ as accurately as we please, by taking n large enough.

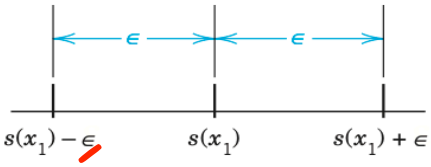


Fig. 104. Inequality (9)

Consider the power series,

$$\text{(1)} \qquad \sum_{m=0}^{\infty} a_m(x - x_0)^m = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \cdots.$$

converges for all x in the interior of the interval, that is, for all x for which

$$\text{(10)} \qquad |x - x_0| < R$$

and diverges for $|x - x_0| > R$. The interval may also be infinite, that is, the series may converge for all x.

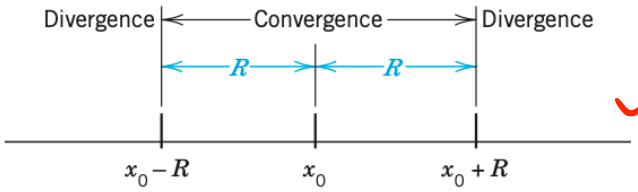


Fig. 105. Convergence interval (10) of a power series with center x_0

The quantity R in Fig. 105 is called the **radius of convergence** (because for a *complex* power series it is the radius of *disk* of convergence). If the series converges for all x, we set $R = \infty$ (and $1/R = 0$).

The radius of convergence can be determined from the coefficients of the series by means of each of the formulas

$$\text{(11)} \qquad \text{(a)} \quad R = 1 \Big/ \lim_{m\to\infty} \sqrt[m]{|a_m|} \qquad \text{(b)} \quad R = 1 \Big/ \lim_{m\to\infty} \left|\frac{a_{m+1}}{a_m}\right|$$

EXAMPLE 4 **Convergence Radius $R = \infty, 1, 0$**

For all three series let $m \to \infty$

$$e^x = \sum_{m=0}^{\infty} \frac{x^m}{m!} = 1 + x + \frac{x^2}{2!} + \cdots, \qquad \left|\frac{a_{m+1}}{a_m}\right| = \frac{1/(m+1)!}{1/m!} = \frac{1}{m+1} \to 0, \qquad R = \infty$$

$$\frac{1}{1-x} = \sum_{m=0}^{\infty} x^m = 1 + x + x^2 + \cdots, \qquad \left|\frac{a_{m+1}}{a_m}\right| = \frac{1}{1} = 1, \qquad R = 1$$

$$\sum_{m=0}^{\infty} m!x^m = 1 + x + 2x^2 + \cdots, \qquad \left|\frac{a_{m+1}}{a_m}\right| = \frac{(m+1)!}{m!} = m + 1 \to \infty, \qquad R = 0.$$

Convergence for all x ($R = \infty$) is the best possible case, convergence in some finite interval the usual, and convergence only at the center ($R = 0$) is useless. ■

THEOREM 1

Existence of Power Series Solutions

If p, q, and r in (12) *are analytic at* $x = x_0$, *then every solution of* (12) *is analytic at* $x = x_0$ *and can thus be represented by a power series in powers of* $x - x_0$ *with radius of convergence* $R > 0$.

consider the ODE

$y''+p(x)y'+q(x)y = r(x)$ → 

(1) has a power series solution if p, q, r are analytic by the above theorem

A function f(x) is said to be analytic at a point x = xo if it can be expanded as a power series in powers of (x - xo) with positive radius of convergence

Q: Determine the radius of convergence

1) $\sum_{n=0}^{\infty} (n+1)n\,x^n$

Here $a_n = (n+1)n$

$$R = \frac{1}{\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|}$$ ✓

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{(n+2)(n+1)}{(n+1)\cdot n}$$

$$= \lim_{n\to\infty}\frac{n^2\left(1+\frac{2}{n}\right)\left(1+\frac{1}{n}\right)}{n^2\left(1+\frac{1}{n}\right)}$$

$$= 1$$

$\therefore R = 1$ interval of convergence $|x|<1$

2) $\sum_{n=0}^{\infty} \frac{(-1)^n}{k^n}\,x^{2n} = \sum_{n=0}^{\infty} \frac{(-1)^n}{k^n}(x^2)^n$

Now, $a_n = \frac{(-1)^n}{k^n}$

$$|a_n| = \frac{1}{|k|^n}$$ ✓

$$R = \frac{1}{\lim_{n\to\infty}\sqrt[n]{|a_n|}}$$

$$\lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}|a_n|^{1/n}$$ ✓

$$= \lim_{n\to\infty}\left(\frac{1}{|k|^n}\right)^{1/n} = \frac{1}{|k|}$$

$$\therefore R = \frac{1}{1/|k|} = |k|$$ ✓

Convergence interval is

$$|x^2| < |k|$$ ✓

$$\therefore |x| < \sqrt{|k|}$$

∴ Radius of convergence $= \sqrt{|k|}$

3) $$\sum_{m=0}^{\infty} \frac{x^{2m+1}}{(2m+1)!} = \sum_{n=1}^{\infty} \frac{x^n}{n!}$$ ✓, $n = 2m+1$

Here $a_n = \frac{1}{n!}$, $a_{n+1} = \frac{1}{(n+1)!}$

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{\frac{1}{(n+1)!}}{\frac{1}{n!}}$$ ✓

$$= \lim_{n\to\infty} \frac{n!}{(n+1)!}$$

$$= \lim_{n\to\infty} \frac{n!}{n!(n+1)}$$

$$= \lim_{n\to\infty} \frac{1}{n+1}$$

$$= 0$$

$$\therefore R = \frac{1}{0} = \infty$$

The above series is convergent for all x and convergence interval is $|x| < \infty$ (entire Real line)